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SUMMARY

Applied AI engineer shipping production LLM and agentic systems into regulated enterprise workflows - automations live across Accounting, Tax, Legal, HR, Digital Marketing, and Asset Management at a 900-employee REIT. Governing design principle: arithmetic and reconciliation stay in deterministic code; the model is confined to extraction and classification. Outcomes include ~1 FTE reduction, ~\$100K use-tax penalty exposure resolved, and the #1 most-used internal GPT company-wide.

WORK EXPERIENCE

Senior AI Analyst (Applied AI Engineering & Enablement) | Welltower Inc., Dallas, TX | Aug 2025 - Present

- Architected the shared orchestration platform that moved six departments from one-off AI experiments to reliable, repeatable production workflows - a queue orchestrator with a reusable sub-workflow library, atomic retry, and centralized error recovery, so onboarding a new department automation requires no orchestrator change.
- Owned each system end to end - discovery and requirements with the department, architecture, build, UAT, deployment, logging, failure monitoring, and iteration - shipping ChatGPT custom GPT, Streamlit applications, and n8n workflow automation into Accounting, Tax, Legal, HR, Digital Marketing, and Asset Management.

Data Scientist (Research Assistant) | Seattle University, Seattle, WA | Sep 2022 - Aug 2025

- Predicted asthma onset at **88%** accuracy - **20%** better than traditional methods - with transfer-learning and ensemble models on sparse medical data, improving forecasting accuracy by **12%** over standard neural networks, on HIPAA-compliant **Docker** and **AWS** pipelines.
- Cut AWS deployment time **65%** across **50+** integrations, replacing 15-20 minutes of manual console work per deployment with CloudFormation templates, then drove adoption to become the university's standard deployment process.
- Built and rolled out a retrieval-augmented generation (RAG) search system over thousands of hours of lecture recordings, adopted by five academic departments - gathered requirements, demoed a proof of concept, then productionized a LangChain and Streamlit interface at 89% precision/recall, cutting latency 35% and compute cost 50%.

Technical Consultant - SaaS Implementation | Yardi Systems, Dubai, UAE | Apr 2019 - Jul 2022

- Cut implementation time and deployment cost **75%** for real estate **BI** modules with **SSIS** and **SQL Server** ETL pipelines, and accelerated customer onboarding **60%** with multi-format ingestion (batch, real-time, historical) partitioned for **40%** faster processing.
- Rescued a BI engagement after a failed CEO demo - audited every KPI back to source, rebuilt all dashboard calculations, and secured written sign-off report by report with the client's managers and analysts, then co-presented to the CEO alongside two of their own managers; the account referred Yardi to its sister companies, unlocking additional contracts.
- Delivered end-to-end BI for **20+** UAE real estate clients - requirements through build, testing, and client-validated UAT - then drove adoption through C-suite and property-staff walkthroughs, turning delivered dashboards into the reporting system leadership ran the business on.

Software Engineer | Yardi Systems, Pune, India | Nov 2016 - Mar 2019

- Retained a \$3M+ account on a twice-failed Commercial Leasing Manager (CRM) implementation at a global asset manager's India retail platform - sent in as the technical lead after the client moved to leave the platform, gathered requirements directly from users on-site, and rebuilt their configuration: lease approval workflows (business process management), automations, reports, and deal and proposal templates. Monthly reported issues fell from ~100 to 7-8, the client signed off inside two months, and system usage rose 50%; promoted to Dubai on the strength of the save.
- Powered real-time dashboards across **100+ KPIs** with a dimensionally modeled data warehouse and data marts - **15%** faster processing via incremental loading - and set the governance standards, automated testing, and query tuning behind them.

PROJECTS (Welltower Inc)

Insurance Requirements Extraction & Compliance Benchmarking - Legal

- Replaced manual clause-by-clause insurance review in acquisition diligence for the Legal team - document-intelligence workflows read purchase and sale agreements and extract every requirement with its dollar limits, then a deterministic benchmark engine flags each limit green, yellow, or red against historical deal statistics.
- Made every extracted limit auditable to its source clause - a three-stage **LLM** chain on **Amazon Bedrock** handling deal context, clause extraction with verbatim quotes and page citations, then normalization of free-text limits onto standard coverage types, each stage grounded with anti-hallucination rules.

Webb - Enterprise HR Assistant

- Cut HR support tickets by **85%** by migrating the company-wide HR assistant from a legacy **Slack** bot to a **ChatGPT custom GPT** - **700** users (**78%** of organization), **200** monthly active users, **100+** daily queries, and the **#1** most-used custom GPT company-wide for six months.
- Held the assistant secure through emerging adversarial use across successive prompt revisions - injection red-flag patterns, identity lock, silent refusal of system-prompt disclosure, web search narrowed to checklist-gated cases, and the connector scoped to user identification.
- Engineered out the silent-wrong-answer failure mode - a location gate withholds answers until the employee's region is known, routing only accurate information, and barring extrapolation from partial documents.

- Shipped every revision defect-free by gating release on a regression suite and a 10-day Human Capital hallucination audit - cases pinning both correct and specific wrong responses across policy, regional routing, refusals, and injection attempts.
- Closed the human-in-the-loop feedback loop on live traffic - an OpenAI Compliance API pipeline landing 100+ daily production conversations in Snowflake behind a review frontend with flagging, turning real usage into a reviewed, flagged dataset and now HR's standing weekly review.

Use Tax Compliance Automation

- Resolved approximately **\$100K** in use tax penalty exposure and established the standing control preventing recurrence across all operating jurisdictions - automating use tax determination for state, city, and county codes for **3,000+** invoices monthly across **1,500+** properties.
- Eliminated a hallucination risk unacceptable in a compliance calculation - rebuilt an **Azure** live-search-per-line-item design as an **n8n** workflow grounded in **Thomson Reuters** tax code data: a reference-data lookup handles known commodities, unmatched lines escalate to a higher-capability model returning a cited proposal, and line-total mismatches route to human review.

Property Site Map - Data-Driven Rent Setting

- Built a custom GPT generating review-ready **Excel** site maps for unit-level rent setting - a controlled A/B test across two structurally identical halves of one property showed generated pricing beat the incumbent on occupancy, rental income, and resident preference; asset management adopted it as the standing pricing framework.

Talk Further Call Center Analytics

- Recovered **15%** additional leads from dropped conversations and drove a **3%** revenue increase across **50+** properties - replaced a third-party batch transcript vendor with an in-house AI analysis system moving **3,000+** calls to daily internal review, now property managers' standing **KPI** via **Power BI** dashboards linking call topics to occupancy and conversion.

Performance View Chat - Natural Language to SQL to Claude Plugin

- Built a natural-language-to-SQL system for the Investment team across **1,500+** properties, removing their dependency on the data team for ad-hoc analysis - later retiring the **Streamlit** UI for a **Claude** plugin pairing a **Snowflake** connector with skills encoding the team's output formats and report structures, so analysis runs directly in chat.

Scope Creation Tool - Capital Projects Scope Generation

- Replaced manual per-project drafting for the capital projects team with a guided generation tool that turns a project manager's answers into complete construction scope-of-work documents - now used by every project manager on the team.
- Kept every generated spec traceable to precedent and correct for its site - grounded generation in a **1,022-item** scope corpus and **16** templates across **41** trade categories, conditioned per trade on a **2,384-property** master dataset, then an automated climate and wind pass rewrites specs for the building's environment.
- Preserved **Word** numbering, styles, and run-level formatting through an **LLM** rewrite - a similarity-based merge re-aligns generated text onto the original paragraph objects and keeps the original wherever confidence is low.

Journal Entry Tie-Out Automation

- Recovered ~1 FTE annually by cutting payroll tie-out from **8** people at **6** hours weekly to a single exception reviewer at **4-5** hours - implementing payroll cash tie-out, payroll journal entries, and suspense item clearing as **n8n** production workflows, after documenting **25-30** undocumented processes across **5-6** departments as machine-readable specifications.
- Designed reconciliation workflows structurally incapable of mutating the general ledger - proposes changes (missing chart-of-accounts entries, mapping corrections) for human approval rather than applying them, making an incorrect automated inference impossible to post.

SKILLS

- **AI & Machine Learning:** Python, PyTorch, RAG, LLM evaluation, embeddings, vector databases (FAISS, Pinecone), prompt engineering, guardrail and anti-hallucination design, OpenAI (GPT, custom GPT, Compliance API), Anthropic Claude (plugins, skills, MCP), Amazon Bedrock, LangChain, Hugging Face, agentic systems, multi-turn workflows, tool use, evaluation harnesses, dataset curation, model fine-tuning.
- **Data & Analytics:** Snowflake (Cortex Analyst), SQL, T-SQL, ETL, SSIS, dimensional modeling, data warehousing
- **Cloud & Infrastructure:** AWS (S3, SageMaker, Transcribe, Bedrock), Docker, Git, GitHub Actions CI/CD, RESTful APIs
- **Applications & Reporting:** Streamlit, n8n, Power BI, Tableau, KPI dashboards, Excel

EDUCATION

MS, Computer Science (Data Science specialization) | Seattle University, Seattle, WA | Sep 2022 - Jun 2024

MS, Computer Applications | Symbiosis International University, India | Jul 2015 - Apr 2018

BS, Information Technology | University of Mumbai, Mumbai, India | Jun 2011 - Jan 2015

PUBLICATIONS & CERTIFICATIONS (Research Papers: github.com/jay-singhvi/publications)

- Published in DAWAK 2024 - [Incremental SMOTE with Control Coefficient for Classifiers in Data Starved Medical Applications](#)
- Published in SAC 2025 - [A Retrieval-Augmented Framework for Meeting Insight Extraction](#)
- Published in IEEE Access 2026 - [Transfer Learning for Deep Feature Extraction Under Extreme Patient-Level Data Scarcity in Health Risk Prediction](#)
- CITI Program - Responsible Conduct of Research - Engineers | Human Subjects Research for IRB (Faculty, Staff, and Student)
- AWS Certified AI Practitioner (AIF-C01) (Other Certificates: linkedin.com/in/jay-singhvi/details/certifications/)